Soft constraint continuation: Cantera vs NN | Test IC 04

T_0 = 1317.5 K P_0 = 1.244 atm ϕ_0 = 1.984 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

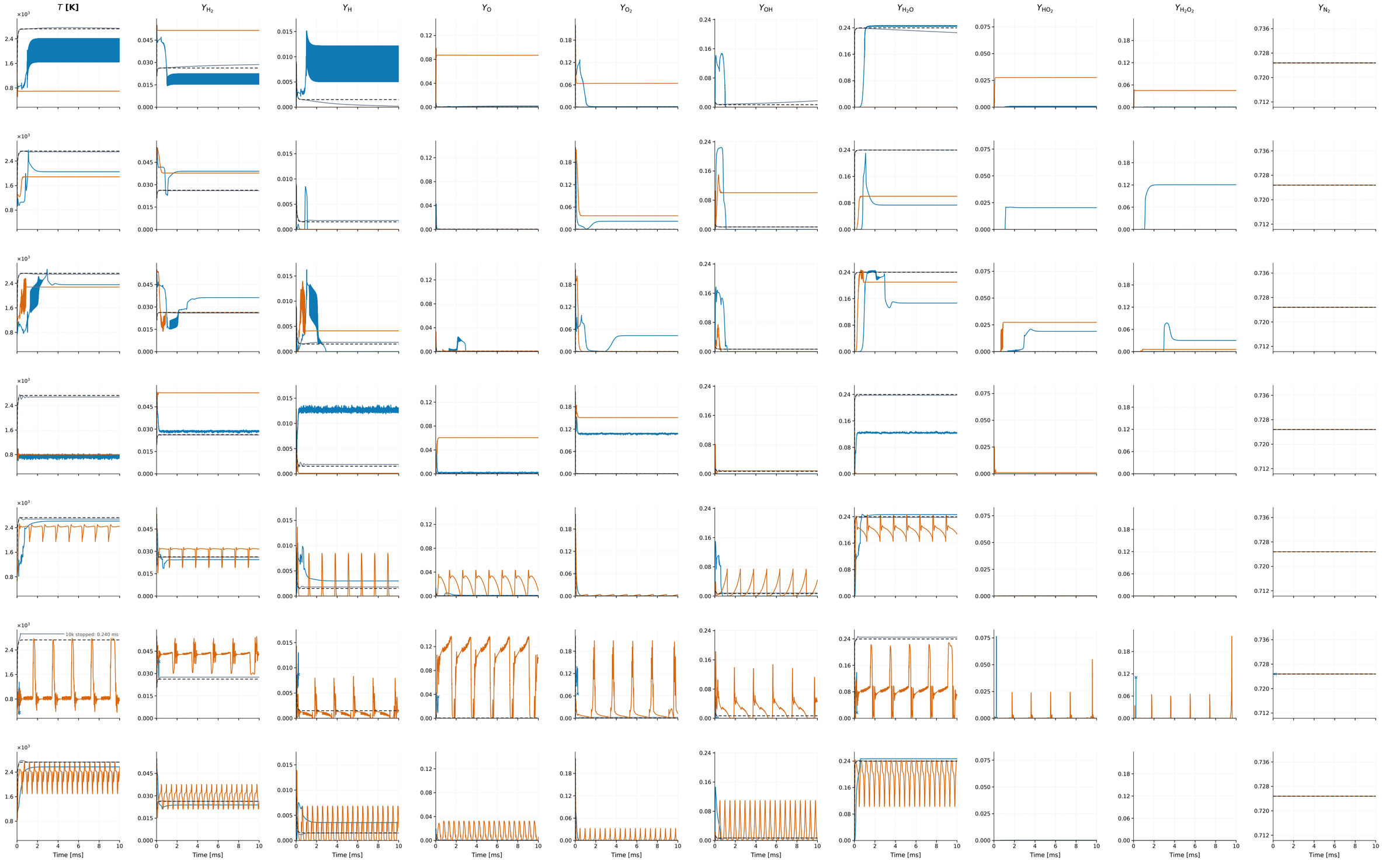
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 04 | NRMSE (util.py)

T0 = 1317.5 K P0 = 1.244 atm phi0 = 1.984 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	0.0108	0.0583	0.593	2.33	0.0953	0.832	0.0354	0.149	0.418	0	Complete
1 ANN10	soft10k	0.346	0.354	4.88	0.747	2.76	5.37	0.288	170	740	0	Complete
1 ANN10	softlong	0.745	0.957	1	153	6.2	1	1	8.51e+03	3.52e+05	0	Complete
2 P-only DeepONet10	baseline	8.74e-03	0.0113	0.165	0.217	0.0395	0.0155	2.22e-03	0.206	0.266	0	Complete
2 P-only DeepONet10	soft10k	0.29	0.486	1.13	3.72	2.47	8.33	0.704	5.9e+03	8.86e+05	0	Complete
2 P-only DeepONet10	softlong	0.317	0.464	0.987	0.955	4.39	13.7	0.602	1	0.79	0	Complete
3 ANN10 + P-only DeepONet10	baseline	9.87e-03	0.0137	0.196	0.258	0.0434	0.031	2.92e-03	0.155	0.2	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.262	0.385	2.17	8.88	4.02	6.24	0.44	4.9e+03	2.39e+05	0	Complete
3 ANN10 + P-only DeepONet10	softlong	0.194	0.173	2	2.21	2.8	1.48	0.197	8.06e+03	4.46e+04	0	Complete
4 T/P DeepONet10	baseline	0.0189	0.0104	0.227	0.392	0.0858	0.218	9.30e-03	0.336	0.28	0	Complete
4 T/P DeepONet10	soft10k	0.727	0.125	7.04	6.23	10.4	1.04	0.489	5.78	0.913	0	Complete
4 T/P DeepONet10	softlong	0.706	1.06	0.892	106	14.6	0.504	0.999	407	0.41	0	Complete
5 ANN10 + T/P DeepONet10	baseline	0.016	8.47e-03	0.186	0.267	0.0804	0.21	8.43e-03	0.233	0.181	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.137	0.11	1.45	2.28	0.847	3.46	0.134	59.7	588	0	Complete
5 ANN10 + T/P DeepONet10	softlong	0.134	0.209	1.4	44	1.19	3.28	0.182	48.7	595	0	Complete
6 T/P/ϕ DeepONet10	baseline	0.0714	0.0485	0.943	0.811	0.196	0.973	0.0223	2.13	23.7	0	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.240 ms**
6 T/P/ϕ DeepONet10	softlong	0.652	0.618	0.846	178	4.48	4.49	0.652	939	1.00e+05	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	4.70e-03	3.41e-03	0.059	0.0529	0.0354	0.0625	2.65e-03	0.16	0.106	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.108	0.127	1.5	1.53	0.832	2.62	0.115	83.1	639	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	softlong	0.181	0.206	1.54	29	1.14	5.74	0.241	22.4	166	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 05

T_0 = 1049.0 K P_0 = 4.666 atm ϕ_0 = 0.674 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

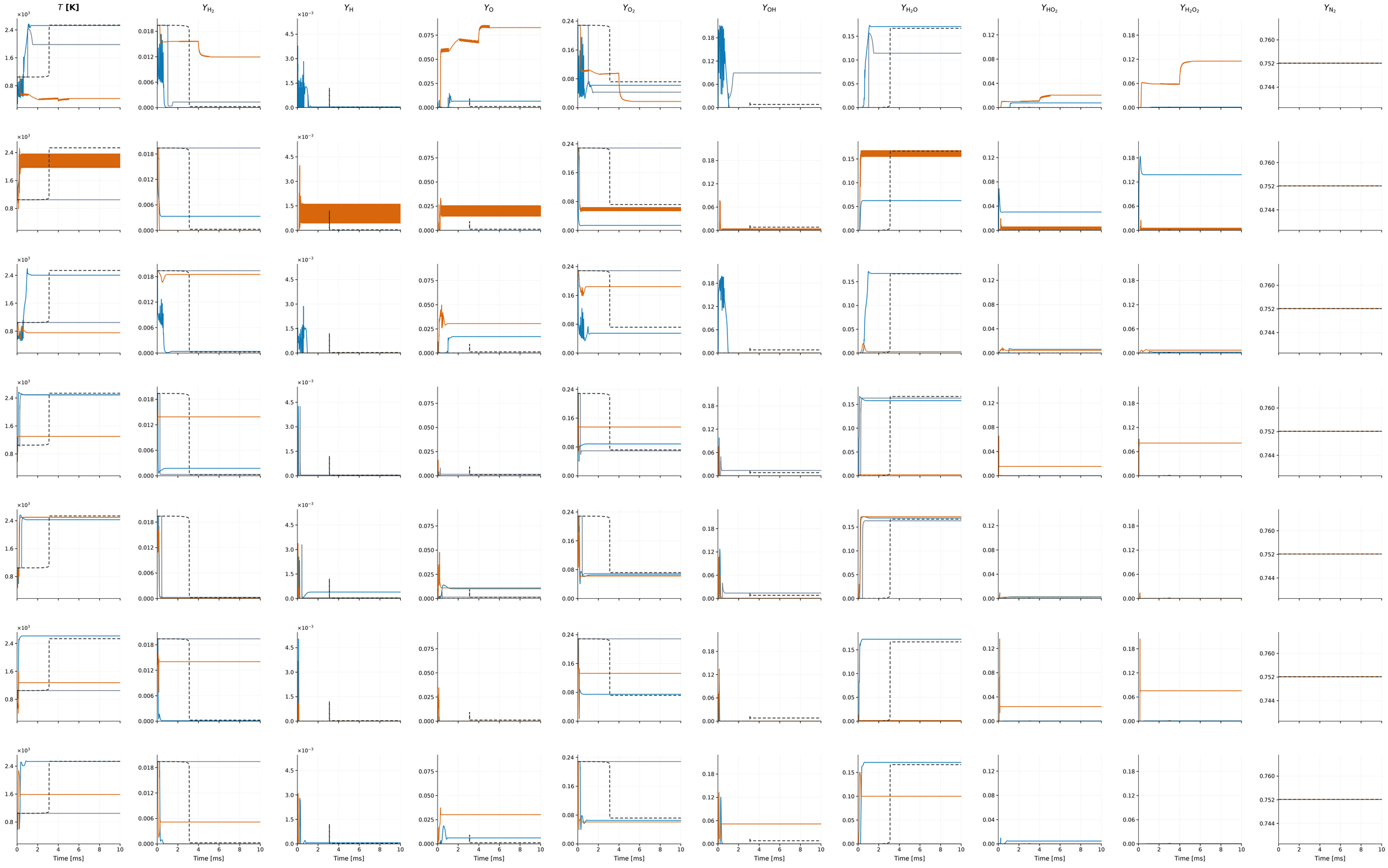
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 05 | NRMSE (util.py)

T0 = 1049.0 K P0 = 4.666 atm phi0 = 0.674 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

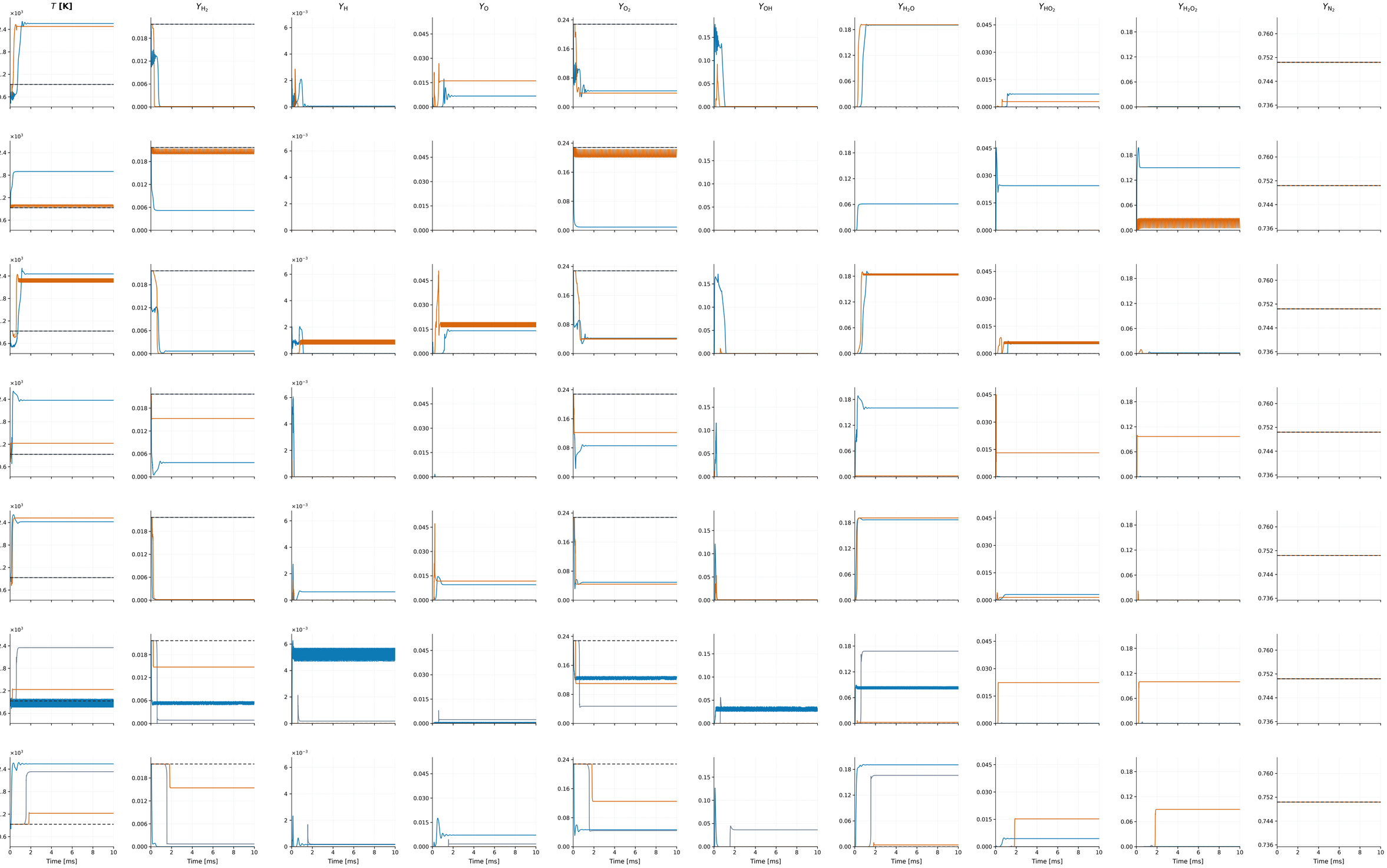
Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	0.299	0.773	1.07	1.02	0.614	11.4	0.509	1.03	1.01	0	Complete
1 ANN10	soft10k	0.318	0.922	10.6	5.47	0.659	6.66	0.585	121	6.62	0	Complete
1 ANN10	softlong	0.813	0.965	1	73.9	0.577	1	1	279	602	0	Complete
2 P-only DeepONet10	baseline	0.565	1.48	1	1	0.929	1	1	0.997	1	0	Complete
2 P-only DeepONet10	soft10k	0.31	0.844	1	1	0.909	1	0.669	507	872	0	Complete
2 P-only DeepONet10	softlong	0.313	0.971	29.1	19.3	0.663	1.08	0.632	60.1	15.9	0	Complete
3 ANN10 + P-only DeepONet10	baseline	0.565	1.48	1	1	0.929	1	1	0.997	1	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.301	0.912	10.3	15	0.687	6.74	0.576	96.3	13.6	0	Complete
3 ANN10 + P-only DeepONet10	softlong	0.681	1.41	1	29.4	0.691	1	0.982	72.1	44.5	0	Complete
4 T/P DeepONet10	baseline	0.35	0.942	2.92	0.887	0.604	1.32	0.626	0.989	0.997	0	Complete
4 T/P DeepONet10	soft10k	0.36	0.909	7.05	1.02	0.56	1.25	0.627	1.03	1	0	Complete
4 T/P DeepONet10	softlong	0.474	1.09	2.47	1.12	0.525	1.09	0.987	255	510	0	Complete
5 ANN10 + T/P DeepONet10	baseline	0.34	0.913	2.09	0.728	0.586	1.34	0.606	0.921	0.996	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.342	0.968	11.8	9.17	0.631	1.8	0.656	42.5	1.31	0	Complete
5 ANN10 + T/P DeepONet10	softlong	0.355	0.964	3.38	10.4	0.644	1.36	0.667	33.8	4.89	0	Complete
6 T/P/ϕ DeepONet10	baseline	0.565	1.48	1	1	0.929	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.388	0.976	8.65	1.03	0.598	1.29	0.675	3.94	3.85	0	Complete
6 T/P/ϕ DeepONet10	softlong	0.482	1.1	3.53	2.42	0.527	1.51	0.991	406	473	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.565	1.48	1	1	0.929	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.354	0.952	6.19	5.75	0.621	1.69	0.651	74.1	3.15	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	softlong	0.388	0.824	4.33	28.8	0.657	6.8	0.568	1.05	0.961	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 07

T_0 = 930.6 K P_0 = 3.722 atm ϕ_0 = 0.754 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s





Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).

Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 07 | NRMSE (util.py)

T0 = 930.6 K P0 = 3.722 atm phi0 = 0.754 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	2.26e-06	1.72e-06	1	360	2.48e-06	1	1	0.531	0.464	0	Complete
1 ANN10	soft10k	1.66	0.968	1.05e+08	8.70e+08	0.791	8.82e+09	1.09e+06	1.38e+04	2.24e+03	0	Complete
1 ANN10	softlong	1.63	0.979	5.31e+07	2.11e+09	0.818	1.71e+09	1.13e+06	5.84e+03	424	0	Complete
2 P-only DeepONet10	baseline	2.84e-06	2.61e-06	1	0.68	4.31e-06	0.986	0.999	0.994	1	0	Complete
2 P-only DeepONet10	soft10k	1.03	0.756	1.40e+05	1.48e+05	0.958	1.20e+05	3.63e+05	5.08e+04	3.81e+05	0	Complete
2 P-only DeepONet10	softlong	0.0691	0.0611	4.75e+03	5.38e+04	0.0922	6.69e+05	108	79.3	5.64e+04	0	Complete
3 ANN10 + P-only DeepONet10	baseline	2.84e-06	2.61e-06	1	0.623	4.31e-06	0.987	0.999	0.992	1	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	1.56	0.942	1.11e+08	1.77e+09	0.802	9.78e+09	1.06e+06	1.04e+04	4.12e+03	0	Complete
3 ANN10 + P-only DeepONet10	softlong	1.39	0.969	2.55e+08	2.50e+09	0.798	1.73e+08	1.07e+06	1.24e+04	3.32e+03	0	Complete
4 T/P DeepONet10	baseline	2.85e-06	2.61e-06	29.5	0.994	4.31e-06	1.56	1	0.993	1	0	Complete
4 T/P DeepONet10	soft10k	1.54	0.832	1.76e+08	1.19e+07	0.628	2.08e+09	9.60e+05	70.1	285	0	Complete
4 T/P DeepONet10	softlong	0.316	0.294	9.38e+06	4.23e+04	0.463	7.44e+07	1.40e+04	2.72e+04	2.44e+05	0	Complete
5 ANN10 + T/P DeepONet10	baseline	2.86e-06	2.49e-06	105	0.972	4.01e-06	14.1	0.99	0.861	0.984	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	1.59	0.993	1.77e+08	1.28e+09	0.781	2.22e+09	1.11e+06	6.2e+03	663	0	Complete
5 ANN10 + T/P DeepONet10	softlong	1.69	0.981	2.37e+07	1.60e+09	0.798	7.70e+08	1.13e+06	3.16e+03	3.35e+03	0	Complete
6 T/P/ϕ DeepONet10	baseline	1.48	0.931	5.41e+07	3.02e+08	0.768	6.67e+09	9.77e+05	107	599	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.0976	0.755	1.48e+09	6.75e+07	0.453	6.92e+09	4.97e+05	2.18	1	0	Complete
6 T/P/ϕ DeepONet10	softlong	0.324	0.315	1.05e+03	145	0.512	6.82e+04	1.57e+04	4.56e+04	2.50e+05	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	1.38	0.888	5.23e+07	2.11e+08	0.74	7.44e+09	9.16e+05	119	900	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	1.71	0.993	6.20e+07	9.94e+08	0.791	2.27e+09	1.13e+06	8.85e+03	1.14e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	softlong	0.284	0.261	66.8	1.44e+03	0.408	2.44e+03	2.30e+04	2.85e+04	2.05e+05	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 18

T_0 = 1069.1 K P_0 = 2.170 atm ϕ_0 = 1.544 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

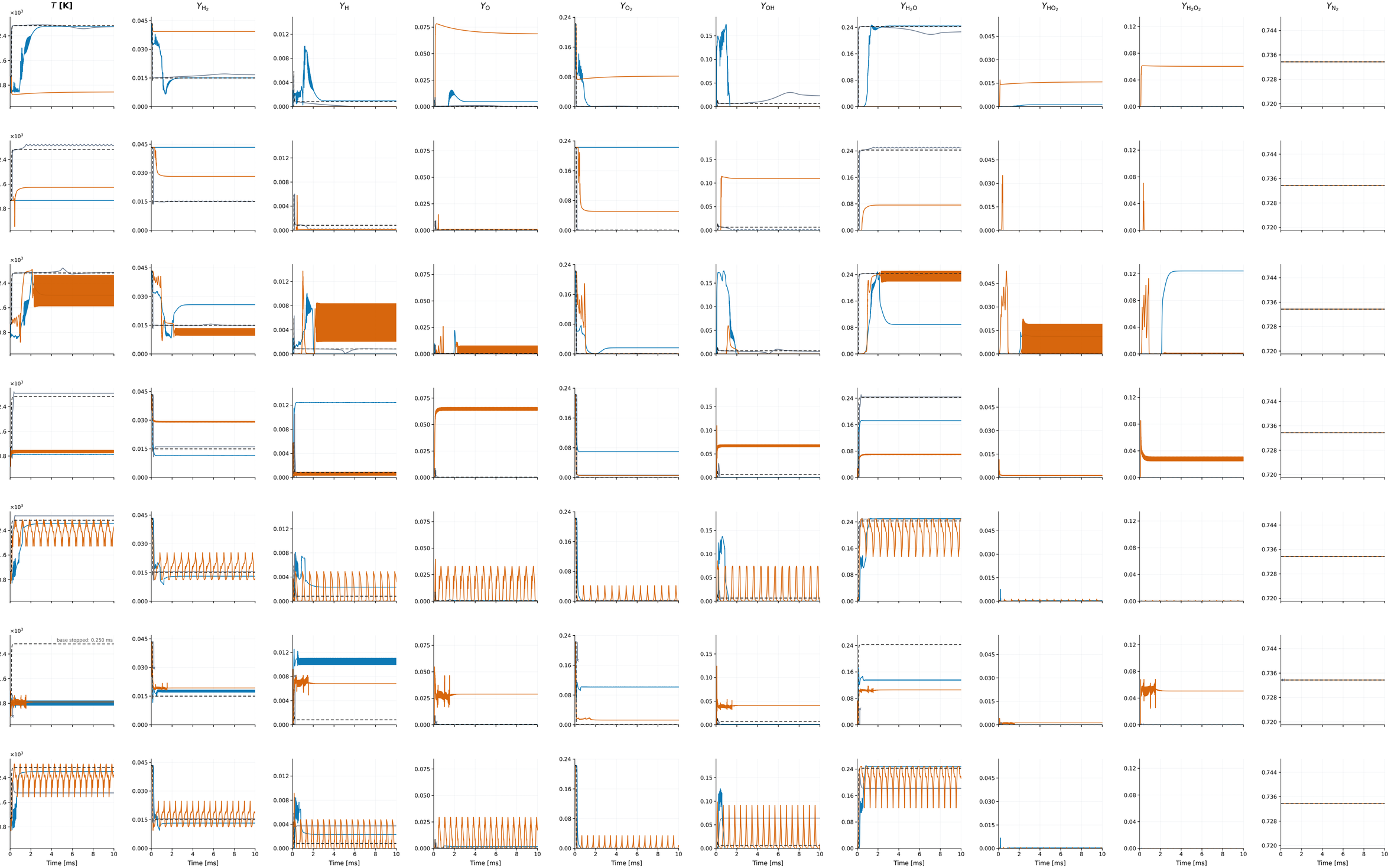
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 18 | NRMSE (util.py)

T0 = 1069.1 K P0 = 2.170 atm phi0 = 1.544 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	0.0255	0.115	0.793	0.959	0.336	1.94	0.0671	1.02	1.19	0	Complete
1 ANN10	soft10k	0.249	0.34	2.24	10	1.06	6.81	0.295	75.5	106	0	Complete
1 ANN10	softlong	0.795	1.54	1	141	3.16	1	1	1.08e+03	5.23e+04	0	Complete
2 P-only DeepONet10	baseline	0.067	0.161	0.806	1.05	0.768	0.82	0.0892	1.54	2.36	0	Complete
2 P-only DeepONet10	soft10k	0.608	1.8	1	1	8.84	1	1	1	1	0	Complete
2 P-only DeepONet10	softlong	0.461	0.889	0.879	1.47	2.52	15.2	0.702	178	2.85e+03	0	Complete
3 ANN10 + P-only DeepONet10	baseline	0.029	0.0992	0.314	0.756	0.462	0.297	0.051	1.31	1.86	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.351	0.681	2.4	4.07	0.985	8.15	0.637	706	9.43e+04	0	Complete
3 ANN10 + P-only DeepONet10	softlong	0.307	0.492	5.82	9.96	1.63	1.43	0.295	1.08e+03	1.59e+04	0	Complete
4 T/P DeepONet10	baseline	0.0652	0.169	1.06	1.06	0.778	1.03	0.0828	1.98	5.67	0	Complete
4 T/P DeepONet10	soft10k	0.687	0.221	13.7	0.996	2.76	0.914	0.293	2.67	0.999	0	Complete
4 T/P DeepONet10	softlong	0.656	0.907	0.42	128	0.483	9.14	0.713	97.3	2.47e+04	0	Complete
5 ANN10 + T/P DeepONet10	baseline	0.0681	0.149	1.03	1.02	0.693	1.02	0.0796	1.62	2.2	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.172	0.24	2.57	0.648	0.895	4.41	0.175	24.9	70.6	0	Complete
5 ANN10 + T/P DeepONet10	softlong	0.158	0.307	2.18	32.3	0.537	4.24	0.195	19.7	160	0	Complete
6 T/P/ϕ DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 0.250 ms**
6 T/P/ϕ DeepONet10	soft10k	0.703	0.185	11.5	0.954	4.01	0.874	0.441	2.19	1	0	Complete
6 T/P/ϕ DeepONet10	softlong	0.693	0.276	7.16	56.4	0.686	5.17	0.567	87	4.45e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.304	0.175	3.44	0.629	0.833	8.52	0.265	2.04	5.77	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.143	0.214	2.3	2.77	0.856	3.11	0.142	28.9	47.4	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	softlong	0.145	0.253	2.18	26	0.46	4.02	0.165	10.6	57.1	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 32

T_0 = 1140.7 K P_0 = 3.354 atm ϕ_0 = 1.463 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

10k stopped: 0.026 ms

long stopped: 0.026 ms

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 32 | NRMSE (util.py)

T0 = 1140.7 K P0 = 3.354 atm phi0 = 1.463 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

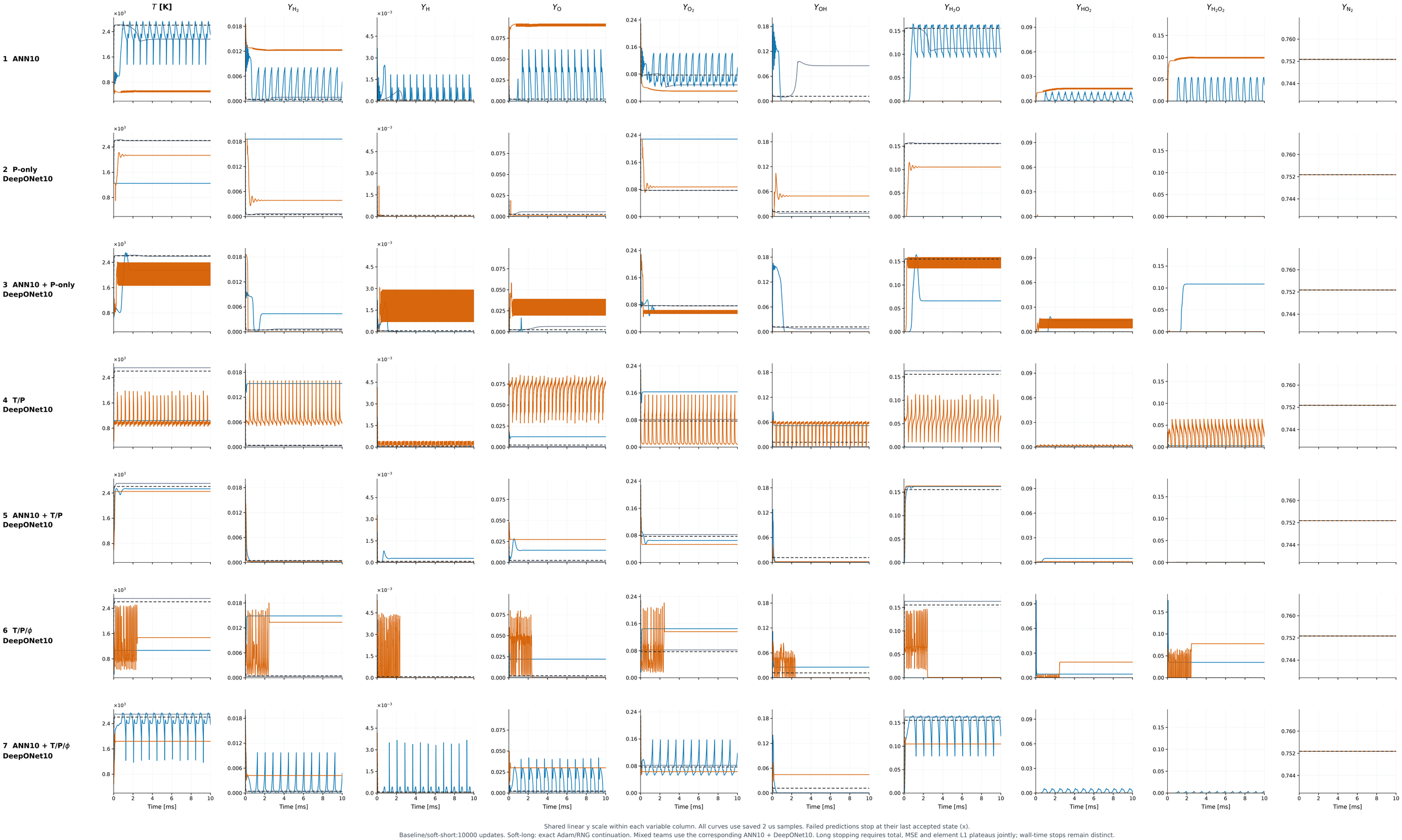
Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	0.0312	0.126	1.48	0.511	0.0533	0.83	0.0317	0.118	1.03	0	Complete
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.026 ms**
1 ANN10	softlong	0.786	1.85	1	114	5.78	1	1	1.49e+03	6.79e+04	0	Complete
2 P-only DeepONet10	baseline	0.0629	0.063	0.97	0.99	0.447	0.972	0.0492	1.24	1.51	0	Complete
2 P-only DeepONet10	soft10k	0.594	2.15	1	1	16.2	1	1	1.05	2.87	0	Complete
2 P-only DeepONet10	softlong	0.16	0.382	0.479	3.59	2.48	4.26	0.241	16.3	8.12e+03	0	Complete
3 ANN10 + P-only DeepONet10	baseline	8.62e-03	0.0401	0.15	0.275	0.276	0.0993	0.0162	0.778	1	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.395	0.466	5.64	13.2	1.17	9.18	0.386	543	1.75e+03	0	Complete
3 ANN10 + P-only DeepONet10	softlong	0.19	0.361	2.79	24.6	2.34	1.49	0.173	148	488	0	Complete
4 T/P DeepONet10	baseline	0.04	0.127	0.968	1	0.844	0.993	0.0346	2.01	13	0	Complete
4 T/P DeepONet10	soft10k	0.657	0.458	12.2	0.996	2.43	8.18	0.341	24.4	1.53	0	Complete
4 T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.026 ms**
5 ANN10 + T/P DeepONet10	baseline	0.0538	0.0417	0.965	0.912	0.184	0.98	0.0292	0.891	0.997	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.162	0.248	3.14	0.602	0.721	3.12	0.142	31.1	91	0	Complete
5 ANN10 + T/P DeepONet10	softlong	0.3	0.743	1.61	11	2.73	7.32	0.462	14.6	98.9	0	Complete
6 T/P/ϕ DeepONet10	baseline	0.285	0.0903	3.13	1.31	0.345	6.75	0.254	2.61	7.38	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.679	0.207	11.1	0.879	3.13	6.86	0.428	5.83	1.02	0	Complete
6 T/P/ϕ DeepONet10	softlong	0.532	1.71	1.01	9.18	8.52	1.22	0.996	2.74e+03	9.24e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.272	0.13	2.39	0.126	0.0193	7.65	0.272	1.92	9.92	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.154	0.173	2.44	8.16	0.604	3.25	0.134	29.5	33.4	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	softlong	0.136	0.288	2.07	17.7	0.793	3.17	0.163	9.07	65.5	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 39

T_0 = 1250.1 K P_0 = 2.297 atm ϕ_0 = 0.646 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).

Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 39 | NRMSE (util.py)

T0 = 1250.1 K P0 = 2.297 atm phi0 = 0.646 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	0.15	0.476	2.07	0.897	0.329	5.49	0.241	0.78	0.837	0	Complete
1 ANN10	soft10k	0.216	4.61	6	8.07	0.391	2.82	0.329	286	2.53e+04	0	Complete
1 ANN10	softlong	0.805	13.5	0.999	34.9	0.594	1	1	802	9.89e+04	0	Complete
2 P-only DeepONet10	baseline	3.93e-03	0.236	0.758	1.17	6.65e-03	0.321	6.61e-03	0.331	0.24	0	Complete
2 P-only DeepONet10	soft10k	0.52	20.5	1	1	1.95	1	1	0.908	0.791	0	Complete
2 P-only DeepONet10	softlong	0.203	5.07	1.11	0.605	0.325	3.25	0.364	4.35	6.43	0	Complete
3 ANN10 + P-only DeepONet10	baseline	5.60e-03	0.233	0.703	1.3	7.30e-03	0.33	6.99e-03	0.292	0.178	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.252	4.86	4.23	1.06	0.29	3.59	0.601	700	1.01e+05	0	Complete
3 ANN10 + P-only DeepONet10	softlong	0.258	3.04	19.9	11.3	0.334	0.999	0.165	541	231	0	Complete
4 T/P DeepONet10	baseline	0.0406	0.212	0.694	0.675	0.0647	0.897	0.0488	0.769	0.794	0	Complete
4 T/P DeepONet10	soft10k	0.6	16.8	0.946	3.92	1.11	3.4	0.999	72.2	3.24e+03	0	Complete
4 T/P DeepONet10	softlong	0.606	7.86	2.05	26.4	0.777	3.57	0.634	99.5	3.62e+04	0	Complete
5 ANN10 + T/P DeepONet10	baseline	0.0362	0.13	0.568	0.562	0.0688	0.891	0.0437	0.833	0.881	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.0613	1.32	2.39	4.95	0.172	1.21	0.0981	253	372	0	Complete
5 ANN10 + T/P DeepONet10	softlong	0.0767	1.17	0.91	9.91	0.311	0.843	0.0804	63	137	0	Complete
6 T/P/ϕ DeepONet10	baseline	0.0408	0.197	0.787	0.562	0.0679	0.955	0.0499	0.634	0.567	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.591	16.2	1.53	7.82	0.873	1.23	1	299	3.62e+04	0	Complete
6 T/P/ϕ DeepONet10	softlong	0.463	13.1	11.9	7.87	0.728	1.6	0.907	904	7.02e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.0351	0.121	0.697	0.309	0.0652	0.934	0.0446	0.686	0.692	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.116	2.76	4.81	6.51	0.303	1.37	0.156	163	1.43e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	softlong	0.296	4.33	1.22	11	0.184	2.79	0.329	3.21	9.85	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 44

T_0 = 1031.1 K P_0 = 2.272 atm ϕ_0 = 0.770 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

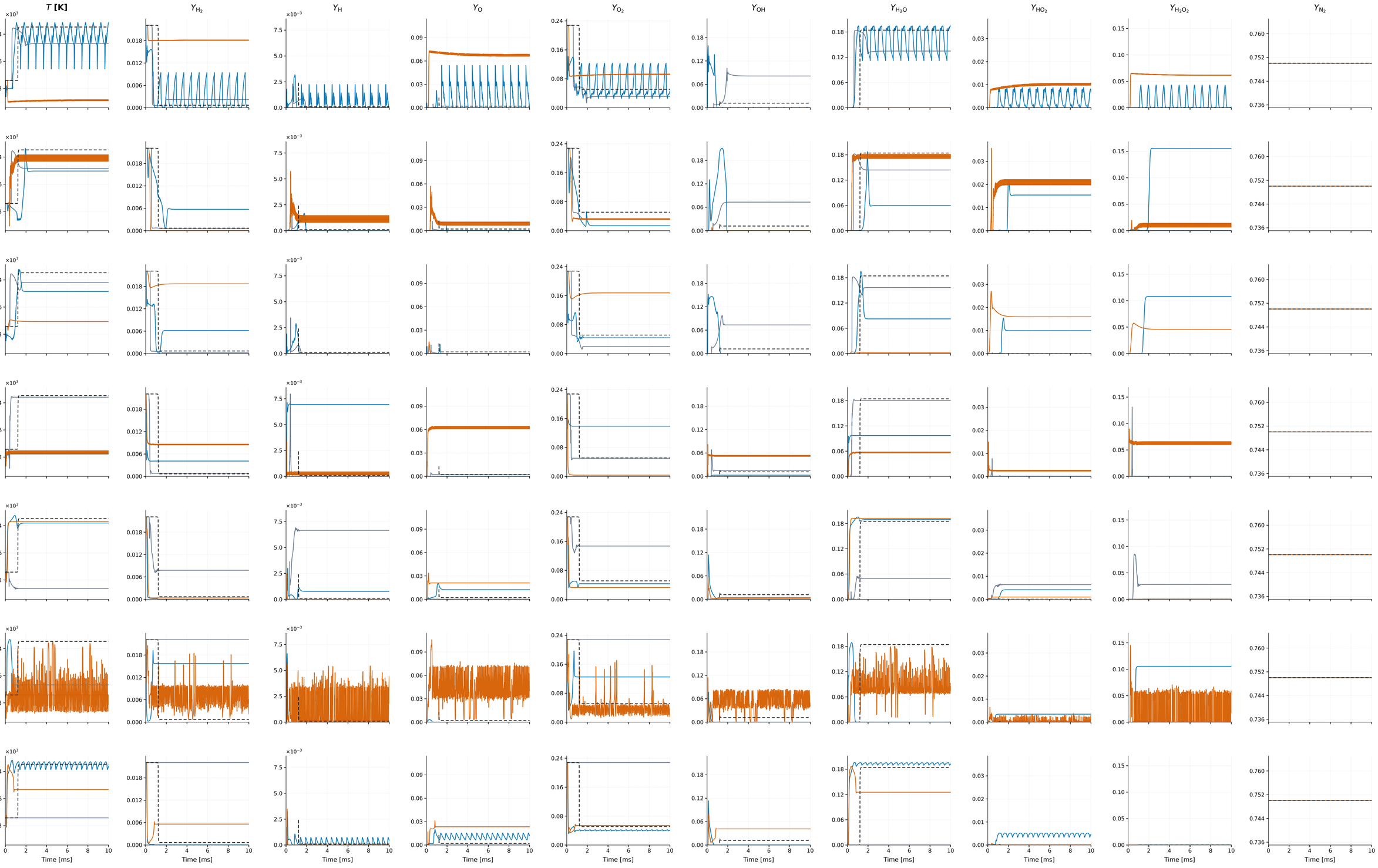
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).

Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 44 | NRMSE (util.py)

T0 = 1031.1 K P0 = 2.272 atm phi0 = 0.770 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	0.231	0.688	2.26	0.999	0.481	5.74	0.363	1.11	1.23	0	Complete
1 ANN10	soft10k	0.191	0.812	6.86	8.51	0.613	2.9	0.287	163	2.13e+03	0	Complete
1 ANN10	softlong	0.824	2.12	1	32.9	0.628	1	1	350	7.61e+03	0	Complete
2 P-only DeepONet10	baseline	0.252	0.701	8.52	0.997	0.528	5.19	0.335	1.82	1.38	0	Complete
2 P-only DeepONet10	soft10k	0.291	0.676	2.81	1	0.519	5.31	0.681	504	1.70e+04	0	Complete
2 P-only DeepONet10	softlong	0.162	0.803	11	4.76	0.616	1	0.283	736	1.32e+03	0	Complete
3 ANN10 + P-only DeepONet10	baseline	0.194	0.773	1.77	1.02	0.628	5.32	0.317	1.03	0.986	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.214	0.867	4.32	1.02	0.538	3.95	0.556	337	1.22e+04	0	Complete
3 ANN10 + P-only DeepONet10	softlong	0.545	2.19	1	1.1	1.2	1	0.989	583	5.66e+03	0	Complete
4 T/P DeepONet10	baseline	0.169	0.784	4.31	0.488	0.547	0.67	0.288	10.8	819	0	Complete
4 T/P DeepONet10	soft10k	0.652	0.881	62.2	0.999	0.943	0.744	0.509	0.93	1	0	Complete
4 T/P DeepONet10	softlong	0.634	1.11	2.14	30.3	0.936	3.85	0.699	91	7.73e+03	0	Complete
5 ANN10 + T/P DeepONet10	baseline	0.782	0.972	57.9	0.965	1	0.638	0.733	221	3.74e+03	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.203	0.927	6.14	5.06	0.655	1.4	0.347	139	39.9	0	Complete
5 ANN10 + T/P DeepONet10	softlong	0.191	0.883	1.32	9.56	0.695	0.858	0.35	34.2	19.1	0	Complete
6 T/P/ϕ DeepONet10	baseline	0.599	2.6	1	1	1.79	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.505	1.94	5.23	1.04	0.896	1.14	1.03	120	1.25e+04	0	Complete
6 T/P/ϕ DeepONet10	softlong	0.586	1.05	15.9	23.2	0.711	4.51	0.575	37.3	3.46e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.599	2.6	1	1	1.79	1	1	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.199	0.937	3.34	4.49	0.672	1.43	0.353	161	45.7	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	softlong	0.323	1.01	1.96	10.6	0.647	2.68	0.438	3.67	2.77	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 45

T_0 = 1181.9 K P_0 = 4.842 atm ϕ_0 = 1.099 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

10k stopped: 0.018 ms

10k stopped: 0.074 ms
long stopped: 0.016 ms

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 45 | NRMSE (util.py)

T0 = 1181.9 K P0 = 4.842 atm phi0 = 1.099 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	8.22e-03	0.379	1.98	1.07	0.214	0.942	0.0629	1.61	5.76	0	Complete
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.018 ms**
1 ANN10	softlong	0.173	0.792	3.78	14.1	1.23	1.41	0.153	117	423	0	Complete
2 P-only DeepONet10	baseline	0.25	1.59	0.904	5.59	2.08	2.48	0.364	2	7.4	0	Complete
2 P-only DeepONet10	soft10k	0.574	1.23	6.93	1.23	1.44	8.57	0.609	16.1	583	0	Complete
2 P-only DeepONet10	softlong	0.314	0.611	5.91	20.6	1.49	0.988	0.146	151	1.79e+03	0	Complete
3 ANN10 + P-only DeepONet10	baseline	0.148	0.982	0.896	1.99	1.19	2.01	0.236	2.46	12.9	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.311	1.05	2.13	15	0.696	4.61	0.35	399	5.68e+03	0	Complete
3 ANN10 + P-only DeepONet10	softlong	0.243	0.367	5	15.4	0.829	1.02	0.109	2.75	38.9	0	Complete
4 T/P DeepONet10	baseline	0.137	1.06	1.85	0.835	2.96	1.05	0.199	4.01	5.37	0	Complete
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.074 ms**
4 T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.016 ms**
5 ANN10 + T/P DeepONet10	baseline	0.0687	0.168	0.958	0.974	0.537	0.996	0.0649	1.19	0.682	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.684	4.53	0.583	12.1	7.96	2.26	1	626	2.30e+04	0	Complete
5 ANN10 + T/P DeepONet10	softlong	0.349	2.36	1.01	11.5	3.81	1.58	0.475	21.3	265	0	Complete
6 T/P/ϕ DeepONet10	baseline	0.0835	0.76	0.851	1.55	2.22	0.889	0.134	2.86	5.55	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.511	4.29	2.03	1.36	8.35	1.52	0.993	750	7.11e+04	0	Complete
6 T/P/ϕ DeepONet10	softlong	0.536	4.58	1.02	2.35	9.14	1.09	0.997	2.07e+03	4.86e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.0594	0.181	0.981	0.969	0.723	0.997	0.0499	0.892	0.616	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.0934	0.457	2.4	5.81	0.839	1.42	0.108	63.1	21	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	softlong	0.343	1.94	0.986	5.25	2.11	4.43	0.496	1.8	11.7	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 53

T_0 = 1257.8 K P_0 = 2.103 atm ϕ_0 = 1.799 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference · Baseline10k (mu=0) · Soft10k (mu=10) · Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

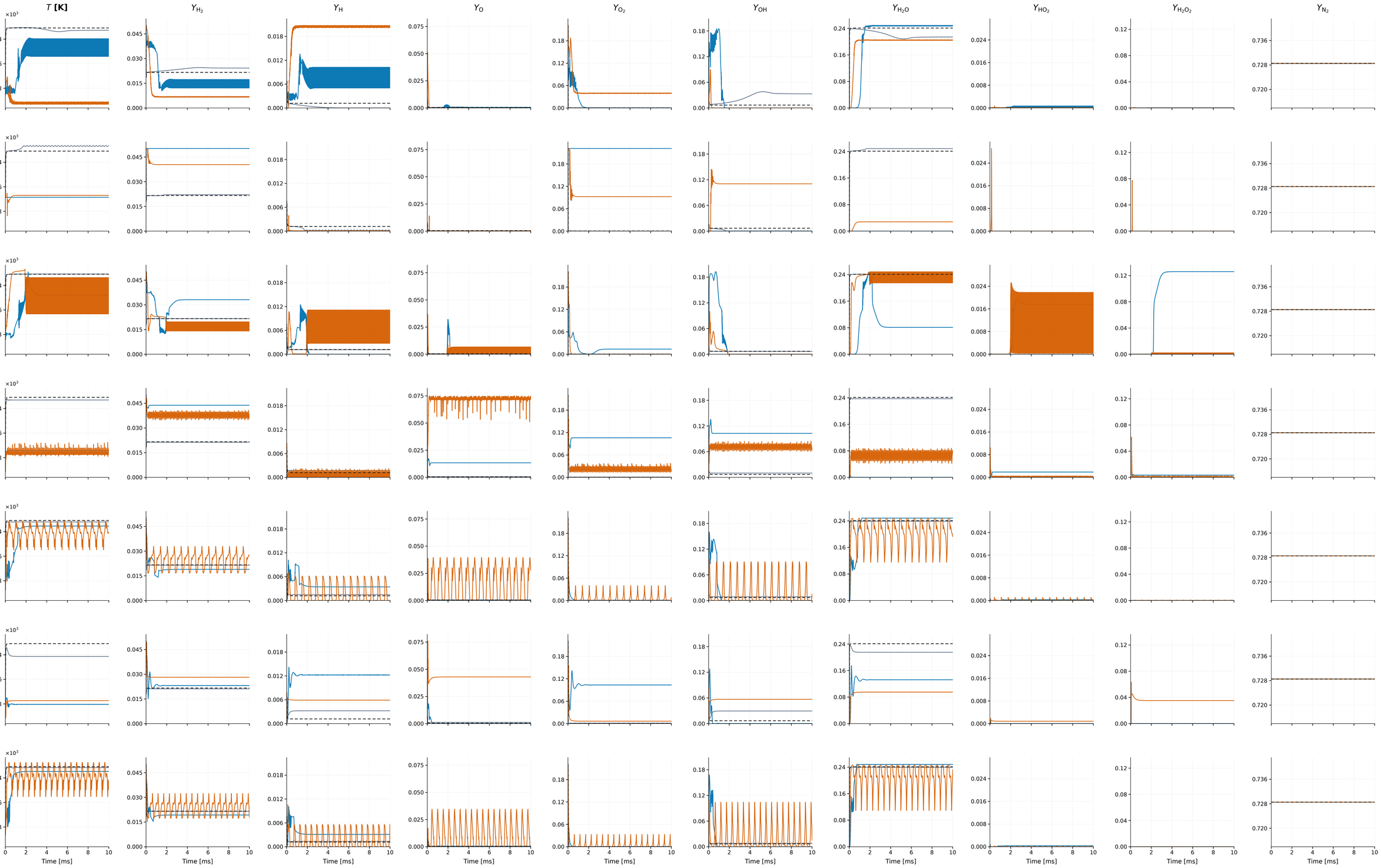
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 53 | NRMSE (util.py)

T0 = 1257.8 K P0 = 2.103 atm phi0 = 1.799 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	0.0249	0.105	0.84	0.642	0.0369	3.25	0.0957	0.118	0.566	0	Complete
1 ANN10	soft10k	0.351	0.399	5.59	0.977	2.4	7.79	0.329	79.8	397	0	Complete
1 ANN10	softlong	0.88	0.692	15.7	6.07	5.28	1.44	0.23	15.2	355	0	Complete
2 P-only DeepONet10	baseline	0.0551	0.0218	0.754	0.845	0.0818	0.868	0.03	0.247	0.641	0	Complete
2 P-only DeepONet10	soft10k	0.545	1.32	1	1	23.3	1	1	0.908	0.929	0	Complete
2 P-only DeepONet10	softlong	0.525	0.879	0.94	1.01	10.1	15.1	0.889	258	1.28e+04	0	Complete
3 ANN10 + P-only DeepONet10	baseline	2.57e-03	2.74e-03	0.0432	0.0484	0.0322	0.0285	1.59e-03	0.131	0.184	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.334	0.512	2.19	6.66	2.08	8.27	0.653	3.08e+03	5.41e+05	0	Complete
3 ANN10 + P-only DeepONet10	softlong	0.312	0.265	5.36	8.63	1.93	1.82	0.139	2.73e+03	6.15e+03	0	Complete
4 T/P DeepONet10	baseline	0.0311	0.0186	0.439	0.467	0.196	0.463	0.0176	0.488	0.557	0	Complete
4 T/P DeepONet10	soft10k	0.628	1.02	0.993	26.2	11.1	14.1	1	369	1.70e+04	0	Complete
4 T/P DeepONet10	softlong	0.65	0.752	0.68	144	2.66	9.52	0.731	110	1.51e+04	0	Complete
5 ANN10 + T/P DeepONet10	baseline	0.0172	0.0109	0.236	0.198	0.13	0.287	0.0107	0.341	0.331	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.185	0.164	2.43	0.491	1.03	5.08	0.175	51.1	427	0	Complete
5 ANN10 + T/P DeepONet10	softlong	0.168	0.257	1.89	41.8	1.41	4.59	0.218	54.7	847	0	Complete
6 T/P/ϕ DeepONet10	baseline	0.149	0.0297	1.67	1.47	0.293	3.28	0.106	1.41	6.57	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.715	0.135	9.05	4.13	10.8	2.04	0.463	14.7	3.79	0	Complete
6 T/P/ϕ DeepONet10	softlong	0.673	0.314	3.9	85.9	1.19	7.3	0.61	167	1.76e+05	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.0109	0.0127	0.15	0.173	0.181	0.196	9.52e-03	0.429	0.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.139	0.145	1.99	0.798	0.952	3.63	0.135	68	375	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	softlong	0.162	0.217	1.75	30.2	1.25	4.94	0.205	18.5	228	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 57

T_0 = 1412.0 K P_0 = 4.083 atm ϕ_0 = 0.677 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

10k stopped: 0.154 ms

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 57 | NRMSE (util.py)

T0 = 1412.0 K P0 = 4.083 atm phi0 = 0.677 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	0.0428	0.666	8.62	0.853	0.176	0.956	0.0319	1.51	4.49	0	Complete
1 ANN10	soft10k	0.178	2.17	4.29	2.86	0.305	3.41	0.261	470	1.22e+03	0	Complete
1 ANN10	softlong	0.106	2.44	2.09	6.79	0.39	1.6	0.184	175	173	0	Complete
2 P-only DeepONet10	baseline	0.0101	0.104	0.933	0.986	0.0348	0.33	0.0267	0.132	1.47	0	Complete
2 P-only DeepONet10	soft10k	0.266	2.27	8.16	8.9	0.588	4.05	0.307	449	9.7e+03	0	Complete
2 P-only DeepONet10	softlong	0.159	4.2	10.6	2.53	0.381	0.934	0.227	207	9.39e+03	0	Complete
3 ANN10 + P-only DeepONet10	baseline	5.19e-03	0.0455	0.906	1.27	0.0366	0.352	0.0232	0.0753	1.05	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.211	2.11	8.65	5.82	0.331	3.19	0.252	376	4.71e+03	0	Complete
3 ANN10 + P-only DeepONet10	softlong	0.1	1.82	3.09	7.05	0.37	0.836	0.105	102	2.18e+03	0	Complete
4 T/P DeepONet10	baseline	3.80e-03	0.0682	0.0852	0.0645	2.64e-03	0.0271	4.83e-03	0.0327	0.023	0	Complete
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.154 ms**
4 T/P DeepONet10	softlong	0.416	16.9	1.24	1.4	1.03	1	0.984	859	3.94e+04	0	Complete
5 ANN10 + T/P DeepONet10	baseline	4.76e-03	0.0786	0.0856	0.0623	1.90e-03	0.045	6.40e-03	0.0294	0.0218	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.0707	2.01	1.01	4.91	0.24	1.16	0.136	49	7.86	0	Complete
5 ANN10 + T/P DeepONet10	softlong	0.0464	1.18	1.15	4.93	0.259	0.936	0.0998	105	104	0	Complete
6 T/P/ϕ DeepONet10	baseline	8.28e-03	0.0376	0.0669	0.036	0.03	0.288	0.0133	0.265	0.371	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.61	19	1.17	11.1	0.915	0.425	1	332	2.39e+04	0	Complete
6 T/P/ϕ DeepONet10	softlong	0.485	8.61	1.62	11.2	0.202	4.45	0.616	0.546	16	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	3.84e-03	0.0728	0.087	0.0721	0.0256	0.213	7.18e-03	0.213	0.325	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.546	12.3	1.25	6.65	0.295	5.28	0.827	3.29	2.89	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	softlong	0.0458	1.01	1.48	4.9	0.272	1.02	0.0994	132	177	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 64

T_0 = 1045.3 K P_0 = 4.301 atm ϕ_0 = 0.595 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).

Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 64 | NRMSE (util.py)

T0 = 1045.3 K P0 = 4.301 atm phi0 = 0.595 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	0.284	0.791	1.02	1.04	0.509	11.4	0.523	1.02	1.01	0	Complete
1 ANN10	soft10k	0.323	0.9	15.3	8.44	0.551	8.46	0.608	86.6	57.5	0	Complete
1 ANN10	softlong	0.785	1.09	1	94.6	0.485	1	1	177	386	0	Complete
2 P-only DeepONet10	baseline	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
2 P-only DeepONet10	soft10k	0.312	0.876	1	1	0.865	1	0.697	451	970	0	Complete
2 P-only DeepONet10	softlong	0.515	1.26	1	1	0.623	0.999	1	0.861	153	0	Complete
3 ANN10 + P-only DeepONet10	baseline	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.29	0.823	15	10.9	0.507	7.59	0.562	91.6	5.43	0	Complete
3 ANN10 + P-only DeepONet10	softlong	0.291	0.858	30.9	32.4	0.54	0.992	0.609	130	10.5	0	Complete
4 T/P DeepONet10	baseline	0.346	0.949	6.56	1.71	0.539	3.79	0.654	1.21	0.992	0	Complete
4 T/P DeepONet10	soft10k	0.381	0.935	9.45	1.06	0.491	1.35	0.694	1.02	0.991	0	Complete
4 T/P DeepONet10	softlong	0.447	0.985	4.08	1.17	0.428	1.18	0.983	231	531	0	Complete
5 ANN10 + T/P DeepONet10	baseline	0.331	0.919	4.92	1.11	0.528	4.59	0.625	1.09	0.99	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.365	0.974	11	9.5	0.541	2.07	0.712	46.6	1.97	0	Complete
5 ANN10 + T/P DeepONet10	softlong	0.371	0.978	4.2	13.1	0.555	1.41	0.724	34.8	0.996	0	Complete
6 T/P/ϕ DeepONet10	baseline	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.399	0.98	11.9	1.12	0.513	1.44	0.728	1.81	1.61	0	Complete
6 T/P/ϕ DeepONet10	softlong	0.455	0.99	6.15	3.26	0.441	2.53	0.989	402	506	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.542	1.37	1	1	0.733	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.379	0.973	7.91	5.32	0.535	2.14	0.715	60.2	2.96	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	softlong	0.385	0.819	7.06	52	0.615	8.5	0.583	1.09	0.954	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 68

T_0 = 1086.3 K P_0 = 3.702 atm ϕ_0 = 1.168 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

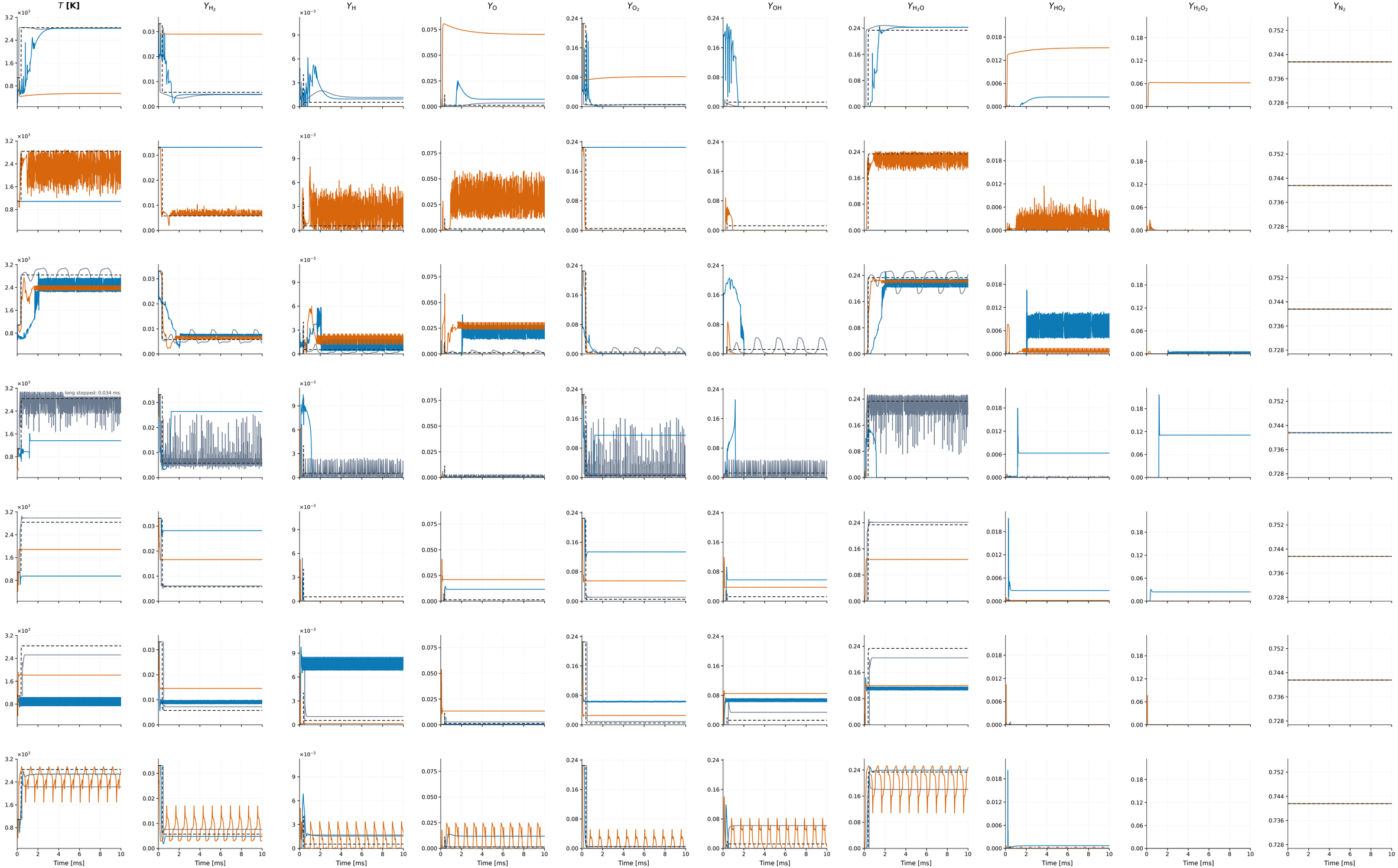
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 68 | NRMSE (util.py)

T0 = 1086.3 K P0 = 3.702 atm phi0 = 1.168 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	0.092	0.499	1.5	1.33	0.738	0.979	0.157	1.47	1.33	0	Complete
1 ANN10	soft10k	0.21	0.484	2.4	5.36	0.82	4.91	0.242	91.8	16.8	0	Complete
1 ANN10	softlong	0.815	2.68	1	51.7	1.75	1	1	622	7.31e+03	0	Complete
2 P-only DeepONet10	baseline	0.615	3.15	1	1	4.94	1	1	0.986	0.989	0	Complete
2 P-only DeepONet10	soft10k	0.615	3.15	1	0.999	4.94	1	1	1.41	4.16	0	Complete
2 P-only DeepONet10	softlong	0.216	0.349	3.52	20.4	0.543	1.25	0.117	94.1	223	0	Complete
3 ANN10 + P-only DeepONet10	baseline	0.0874	0.309	0.756	0.913	0.368	1.32	0.126	1.17	1.26	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.29	0.532	2.38	14.1	0.71	5.79	0.312	278	436	0	Complete
3 ANN10 + P-only DeepONet10	softlong	0.172	0.182	3.09	18	0.245	1.2	0.0739	54.7	92.4	0	Complete
4 T/P DeepONet10	baseline	0.124	0.51	1.4	0.786	0.778	1.36	0.174	2.17	1.35	0	Complete
4 T/P DeepONet10	soft10k	0.53	2.31	4.6	1	2.39	2.59	0.971	253	1.23e+04	0	Complete
4 T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.034 ms**
5 ANN10 + T/P DeepONet10	baseline	0.0776	0.325	1.01	0.993	0.528	1.02	0.102	1.28	1.09	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.662	2.58	1.01	7.32	2.89	3.78	1	118	2.82e+03	0	Complete
5 ANN10 + T/P DeepONet10	softlong	0.338	1.29	1.06	14.4	1.3	2.26	0.461	7.84	2.96	0	Complete
6 T/P/ϕ DeepONet10	baseline	0.132	0.389	0.981	1.04	0.557	1.8	0.167	1.73	1.92	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.688	0.614	13.2	0.704	1.44	4.55	0.531	1.21	1	0	Complete
6 T/P/ϕ DeepONet10	softlong	0.362	1.08	0.79	8.98	0.915	6.1	0.495	6.51	205	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.219	0.369	2.06	0.265	0.472	4.04	0.245	1.75	1.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.0665	0.326	2.4	7.57	0.48	1.38	0.0599	34.2	3.36	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	softlong	0.157	0.725	2.16	7.88	0.903	2.33	0.245	6.73	6.17	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 69

T_0 = 935.0 K P_0 = 2.045 atm ϕ_0 = 0.523 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

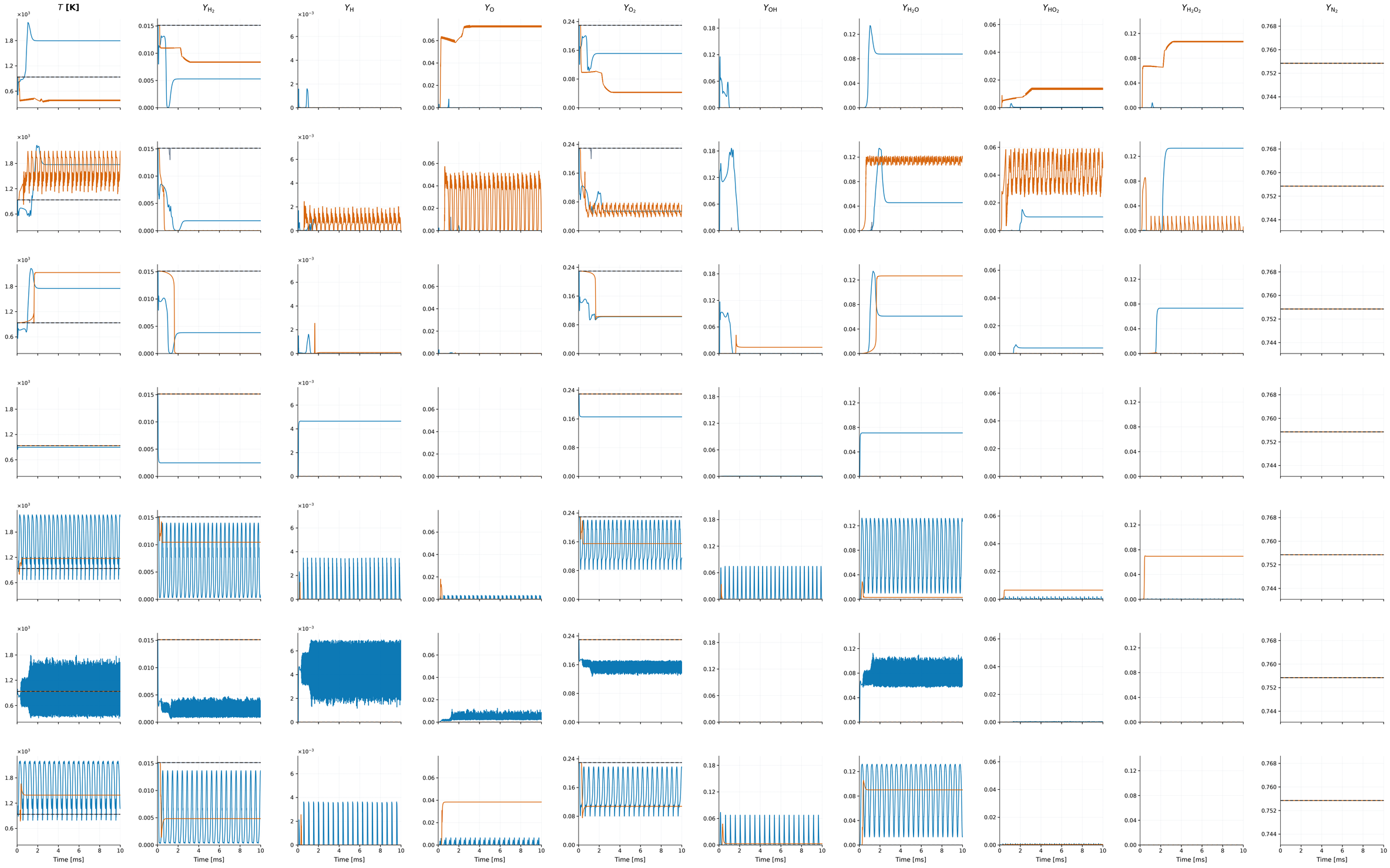
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).

Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 69 | NRMSE (util.py)

T0 = 935.0 K P0 = 2.045 atm phi0 = 0.523 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

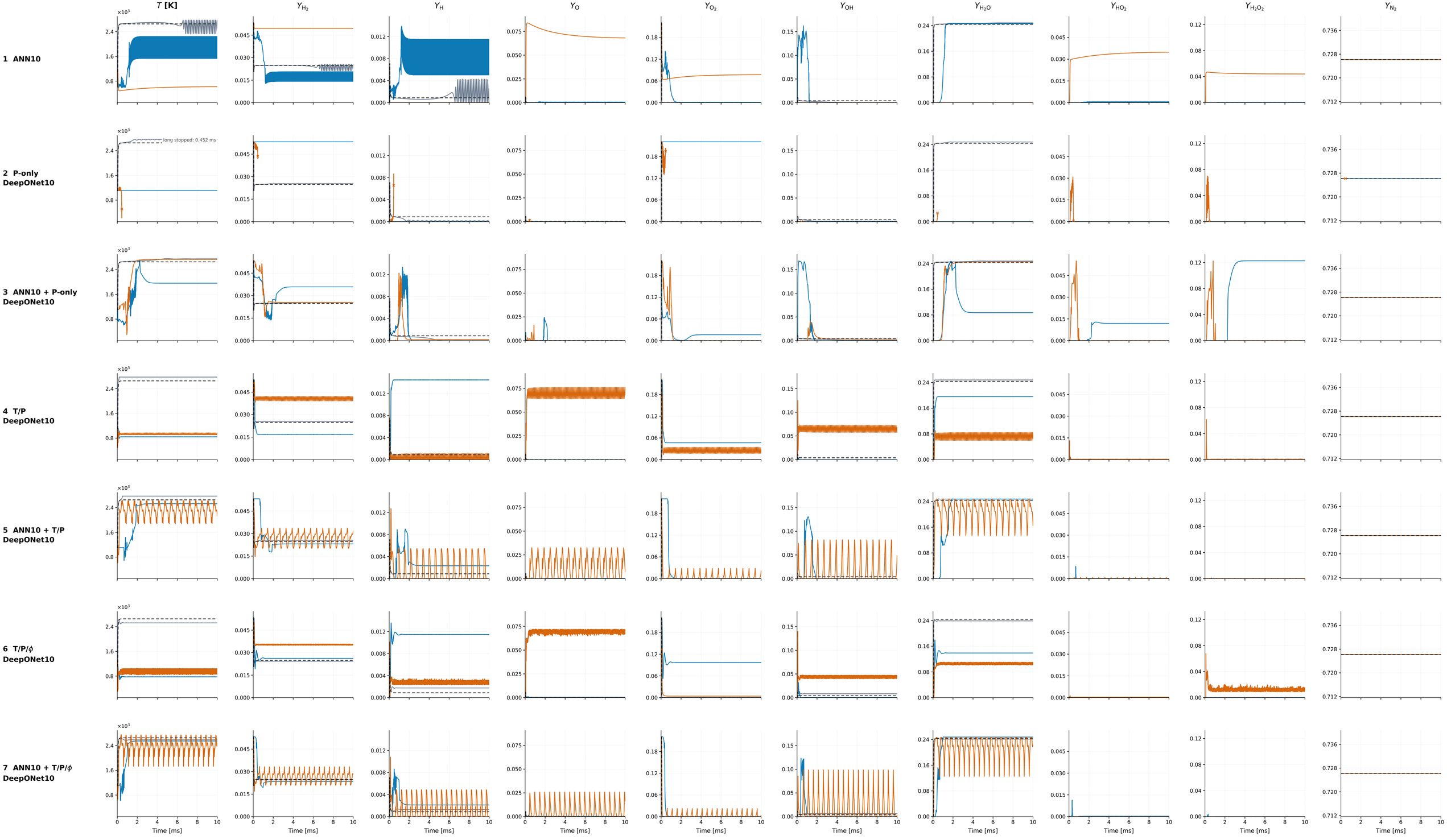
Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	9.88e-07	1.28e-06	1	185	1.53e-06	1	1	0.967	0.997	0	Complete
1 ANN10	soft10k	0.896	0.644	5.90e+07	4.44e+07	0.343	4.06e+09	1.28e+06	2.22e+03	8.1e+03	0	Complete
1 ANN10	softlong	0.584	0.408	6.73e+05	7.33e+09	0.754	1.13e+06	193	5.40e+04	1.11e+06	0	Complete
2 P-only DeepONet10	baseline	2.83e-03	8.45e-03	9.87e+06	5.45e+07	6.11e-03	7.94e+07	1.15e+04	139	52.9	0	Complete
2 P-only DeepONet10	soft10k	0.852	0.854	6.02e+07	4.02e+07	0.73	1.60e+10	7.59e+05	3.97e+04	1.33e+06	0	Complete
2 P-only DeepONet10	softlong	0.784	0.971	2.78e+08	3.34e+09	0.736	5.54e+03	1.66e+06	1.87e+05	1.85e+05	0	Complete
3 ANN10 + P-only DeepONet10	baseline	9.70e-07	1.29e-06	1	0.492	1.55e-06	0.998	0.999	0.988	0.999	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.85	0.735	7.06e+07	1.57e+07	0.538	7.93e+09	9.39e+05	1.69e+04	7.63e+05	0	Complete
3 ANN10 + P-only DeepONet10	softlong	1.16	0.913	4.67e+07	1.00e+07	0.502	3.69e+09	1.73e+06	140	2.95e+03	0	Complete
4 T/P DeepONet10	baseline	9.45e-07	1.27e-06	1	2.28	1.55e-06	0.952	0.963	0.997	1	0	Complete
4 T/P DeepONet10	soft10k	0.0397	0.832	1.48e+09	1.25e+05	0.276	1.91e+08	1.06e+06	59.3	1	0	Complete
4 T/P DeepONet10	softlong	4.29e-04	1.77e-04	8.11e+05	18.8	4.94e-06	5.05e+04	19.5	0.941	1	0	Complete
5 ANN10 + T/P DeepONet10	baseline	3.86e-04	5.04e-04	1	13	5.70e-04	0.999	8.88	94.2	1.33e+03	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.872	0.714	3.22e+08	1.13e+08	0.393	6.85e+09	1.31e+06	1.85e+03	2.89e+03	0	Complete
5 ANN10 + T/P DeepONet10	softlong	0.252	0.303	1.76e+07	1.53e+08	0.318	4.77e+08	6.30e+04	2.86e+04	7.74e+05	0	Complete
6 T/P/ϕ DeepONet10	baseline	2.78e-05	2.32e-05	87.3	109	1.84e-05	399	35.7	7.04	8.79	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.474	0.857	1.49e+09	5.38e+08	0.322	6.67e+06	1.17e+06	476	0.977	0	Complete
6 T/P/ϕ DeepONet10	softlong	3.43e-05	7.56e-05	1.16e+05	243	2.95e-05	1.64e+05	104	1.28	0.999	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	9.40e-06	8.94e-06	18.9	16.9	8.37e-06	52.1	11.3	4.03	5.05	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.934	0.757	3.44e+08	2.12e+08	0.411	5.37e+09	1.41e+06	1.49e+03	1.26e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	softlong	0.481	0.671	4.55e+07	3.97e+09	0.523	1.34e+09	1.32e+06	1.68e+03	455	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 73

T_0 = 1106.0 K P_0 = 1.911 atm ϕ_0 = 1.915 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).

Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 73 | NRMSE (util.py)

T0 = 1106.0 K P0 = 1.911 atm phi0 = 1.915 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	0.0381	0.0487	1.14	0.997	0.157	0.82	0.0187	0.373	6.14	0	Complete
1 ANN10	soft10k	0.384	0.371	8.39	1.47	1.57	10.4	0.302	36.7	176	0	Complete
1 ANN10	softlong	0.779	0.983	1	297	4.16	1	1	3.28e+03	9.55e+04	0	Complete
2 P-only DeepONet10	baseline	0.0376	0.025	0.757	0.734	0.154	0.77	0.019	0.37	0.648	0	Complete
2 P-only DeepONet10	soft10k	0.582	1.13	1	1	12.3	1	1	1	1	0	Complete
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.452 ms**
3 ANN10 + P-only DeepONet10	baseline	0.03	0.0143	0.592	0.58	7.95e-03	0.633	0.0124	0.0794	0.188	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.344	0.437	2.8	14.4	1.41	12.3	0.636	1.04e+03	2.27e+05	0	Complete
3 ANN10 + P-only DeepONet10	softlong	0.195	0.288	1.87	4.76	2.5	1.66	0.303	1.01e+03	4.19e+04	0	Complete
4 T/P DeepONet10	baseline	0.0466	0.0258	0.956	0.772	0.0161	0.986	0.0171	0.459	6.48	0	Complete
4 T/P DeepONet10	soft10k	0.681	0.331	14.7	0.991	2.85	0.99	0.218	8.18	0.969	0	Complete
4 T/P DeepONet10	softlong	0.649	0.638	0.733	285	1.54	15.1	0.705	54.7	5.31e+03	0	Complete
5 ANN10 + T/P DeepONet10	baseline	0.0444	0.0276	0.93	0.673	0.0709	0.961	0.0159	0.422	2.4	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.229	0.305	2.55	1	3.17	7.79	0.299	28.8	167	0	Complete
5 ANN10 + T/P DeepONet10	softlong	0.16	0.19	2.36	65.7	0.835	6.96	0.193	22.3	299	0	Complete
6 T/P/ϕ DeepONet10	baseline	0.0497	0.0333	0.96	1.13	0.242	1.05	0.0279	0.634	0.879	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.706	0.107	11.4	0.899	5.48	1.01	0.437	6.21	1.13	0	Complete
6 T/P/ϕ DeepONet10	softlong	0.646	0.421	2.12	285	0.466	10.1	0.572	23.3	2.87e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.0248	0.0245	0.472	0.561	0.21	0.58	0.0195	0.528	0.672	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.159	0.204	2.05	0.985	2.09	4.76	0.196	41	414	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	softlong	0.143	0.148	1.99	41.8	0.5	7.67	0.173	6.43	79.5	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 74

T_0 = 1222.9 K P_0 = 4.971 atm ϕ_0 = 1.766 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

10k stopped: 0.016 ms
long stopped: 0.250 ms

long stopped: 0.012 ms

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 74 | NRMSE (util.py)

T0 = 1222.9 K P0 = 4.971 atm phi0 = 1.766 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	0.0331	0.0696	1.38	0.728	0.0408	0.804	0.0201	0.211	1.99	0	Complete
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.016 ms**
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.250 ms**
2 P-only DeepONet10	baseline	0.0501	0.0284	1.01	1.17	0.177	1.1	0.0287	0.692	1.41	0	Complete
2 P-only DeepONet10	soft10k	0.469	0.272	7.4	3.92	2.78	14.3	0.386	13.6	734	0	Complete
2 P-only DeepONet10	softlong	0.167	0.16	1.04	4.51	2.13	7.92	0.226	13.4	8.34e+03	0	Complete
3 ANN10 + P-only DeepONet10	baseline	0.045	0.0562	1.06	2.06	0.212	1.67	0.0447	0.662	3.9	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.447	0.277	6.44	1.57	2.51	15.6	0.411	1.89	78.6	0	Complete
3 ANN10 + P-only DeepONet10	softlong	0.157	0.233	1.44	106	1.14	2.25	0.169	25.1	855	0	Complete
4 T/P DeepONet10	baseline	0.04	0.0532	0.97	1.09	0.72	1	0.0224	1.15	9.51	0	Complete
4 T/P DeepONet10	soft10k	0.488	1.06	1	3.99	14.1	1.21	1	1.53e+03	1.34e+05	0	Complete
4 T/P DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.012 ms**
5 ANN10 + T/P DeepONet10	baseline	8.40e-03	0.0292	0.178	0.524	0.521	0.199	0.0177	1.07	0.981	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.676	1.09	0.163	99.1	13.9	7.54	1	1.41e+03	3.82e+04	0	Complete
5 ANN10 + T/P DeepONet10	softlong	0.167	0.242	1.35	30.2	2.21	6.88	0.244	78.7	936	0	Complete
6 T/P/ϕ DeepONet10	baseline	0.155	0.0323	2.06	2	0.572	5.48	0.131	1.5	5.07	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.486	1.04	0.995	7	13.2	1.38	1	2.04e+03	1.39e+05	0	Complete
6 T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.012 ms**
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.152	0.0322	1.86	0.606	0.517	5.99	0.138	1.4	6.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.751	0.308	6.79	1.02	3.79	27	0.751	52.2	45.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	softlong	0.116	0.141	1.51	16.8	1.06	4.94	0.148	4.38	86.3	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 76

T_0 = 1523.6 K P_0 = 3.626 atm ϕ_0 = 1.349 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

10k stopped: 0.016 ms

long stopped: 0.004 ms

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 76 | NRMSE (util.py)

T0 = 1523.6 K P0 = 3.626 atm phi0 = 1.349 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	7.28e-03	0.157	0.733	1.66	0.164	0.967	0.0661	3.08	10.3	0	Complete
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.016 ms**
1 ANN10	softlong	0.223	0.619	3.07	15.3	2.08	1.85	0.204	199	667	0	Complete
2 P-only DeepONet10	baseline	0.0606	0.124	0.537	0.631	0.585	0.637	0.075	1.43	3.56	0	Complete
2 P-only DeepONet10	soft10k	0.567	0.463	3.76	1.08	2.36	7.31	0.609	19.4	427	0	Complete
2 P-only DeepONet10	softlong	0.203	0.571	0.669	0.781	5.49	3.17	0.353	64.8	1.35e+04	0	Complete
3 ANN10 + P-only DeepONet10	baseline	0.0447	0.119	0.469	0.535	0.487	0.537	0.062	1.43	4.32	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.38	0.524	3.64	4.15	1.73	4.74	0.406	1.59e+03	6.13e+03	0	Complete
3 ANN10 + P-only DeepONet10	softlong	0.189	0.28	1.56	12.2	1.79	1.36	0.137	252	92.7	0	Complete
4 T/P DeepONet10	baseline	0.101	0.138	1.02	0.564	0.519	1.13	0.102	4.47	2.83	0	Complete
4 T/P DeepONet10	soft10k	0.646	2.16	0.995	18	14	2.33	1	783	2.67e+04	0	Complete
4 T/P DeepONet10	softlong	0.662	1.74	0.721	21.1	2.51	3.47	0.999	2.67e+03	7.80e+04	0	Complete
5 ANN10 + T/P DeepONet10	baseline	0.0593	0.113	0.713	0.206	0.165	0.838	0.067	0.92	1.56	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.12	0.253	1.38	7.42	1.72	1.42	0.116	44	52.4	0	Complete
5 ANN10 + T/P DeepONet10	softlong	0.356	1.28	1	5.97	8.35	3.32	0.621	22.4	37.7	0	Complete
6 T/P/ϕ DeepONet10	baseline	0.0794	0.0678	0.632	0.485	0.728	0.848	0.0779	2.08	2.75	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.643	2.2	1	18.7	14.9	1.61	1	1.13e+03	3.05e+04	0	Complete
6 T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.004 ms**
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.0672	0.0325	0.52	0.132	0.106	0.894	0.0693	0.916	1.18	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.119	0.229	1.5	5.03	1.75	1.37	0.113	108	56.1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	softlong	0.146	0.514	1.54	8.13	2.16	1.95	0.227	26.7	82	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 78

T_0 = 962.3 K P_0 = 2.443 atm ϕ_0 = 1.140 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

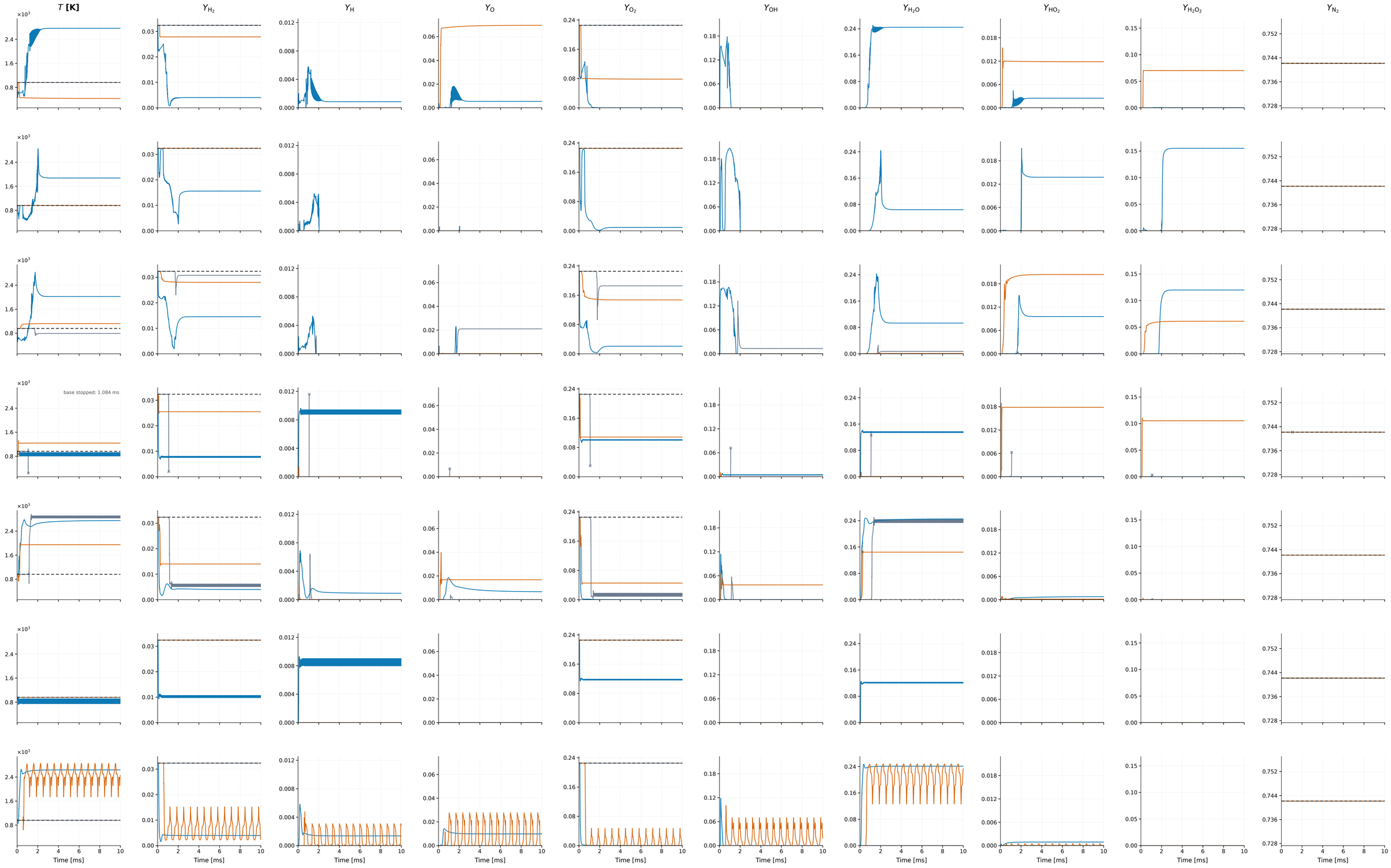
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 78 | NRMSE (util.py)

T0 = 962.3 K P0 = 2.443 atm phi0 = 1.140 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	4.72e-05	2.63e-05	3.24	4.29e+03	6.52e-05	1	1	1.07	2.06	0	Complete
1 ANN10	soft10k	1.75	0.844	7.46e+06	2.11e+07	0.97	2.88e+08	6.18e+04	480	34.4	0	Complete
1 ANN10	softlong	0.54	0.138	9.98e+03	2.39e+08	0.643	1.64e+05	11.5	2.5e+03	1.49e+04	0	Complete
2 P-only DeepONet10	baseline	3.54e-05	1.95e-05	1	0.876	4.00e-05	0.999	0.869	0.848	0.718	0	Complete
2 P-only DeepONet10	soft10k	0.881	0.522	5.79e+06	6.43e+05	0.93	4.64e+08	1.83e+04	2.63e+03	2.96e+04	0	Complete
2 P-only DeepONet10	softlong	4.81e-04	2.22e-04	1	1	3.47e-04	4.89e+03	10.4	0.983	10.5	0	Complete
3 ANN10 + P-only DeepONet10	baseline	0.16	0.0496	8.47e+05	6.63e+07	0.166	1.05e+08	1.77e+03	16.5	4.06	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	1.06	0.561	5.41e+06	6.36e+06	0.894	3.85e+08	2.64e+04	1.89e+03	2.31e+04	0	Complete
3 ANN10 + P-only DeepONet10	softlong	0.162	0.13	379	6.78e+05	0.337	1	235	4.21e+03	1.27e+04	0	Complete
4 T/P DeepONet10	baseline	—	—	—	—	—	—	—	—	—	—	**Stop 1.084 ms**
4 T/P DeepONet10	soft10k	0.112	0.756	4.75e+07	3.2e+03	0.552	3.23e+07	3.61e+04	2.57	1	0	Complete
4 T/P DeepONet10	softlong	0.283	0.211	2.17e+05	8.45e+04	0.516	2.60e+06	239	3.77e+03	2.25e+04	0	Complete
5 ANN10 + T/P DeepONet10	baseline	1.86	0.777	1.64e+06	1.13e+06	0.886	3.31e+07	5.98e+04	8.67	37.4	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	1.79	0.868	7.14e+06	2.99e+07	0.992	8.47e+07	6.41e+04	137	11.4	0	Complete
5 ANN10 + T/P DeepONet10	softlong	1.01	0.56	4.00e+05	5.87e+07	0.789	2.56e+08	3.80e+04	33.9	31.5	0	Complete
6 T/P/ϕ DeepONet10	baseline	4.54e-05	2.59e-05	1	0.884	5.42e-05	0.997	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.156	0.681	4.44e+07	8.67e+04	0.477	7.00e+04	3.23e+04	0.726	1	0	Complete
6 T/P/ϕ DeepONet10	softlong	4.23e-05	2.39e-05	60.9	0.754	5.20e-05	502	0.886	0.994	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	4.54e-05	2.58e-05	1	0.819	5.42e-05	0.995	0.999	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	1.71	0.872	8.09e+06	3.49e+07	0.991	7.41e+07	6.36e+04	186	7.66	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	softlong	1.52	0.783	7.71e+06	4.58e+07	0.933	1.97e+08	5.55e+04	36.6	9.06	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.

Soft constraint continuation: Cantera vs NN | Test IC 88

T_0 = 1502.1 K P_0 = 4.980 atm ϕ_0 = 0.859 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (mu=0) Soft10k (mu=10) Soft long (mu=10)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

long stopped: 0.004 ms

long stopped: 8.002 ms

10k stopped: 0.164 ms

long stopped: 0.004 ms

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
Baseline/soft-short:10000 updates. Soft-long: exact Adam/RNG continuation. Mixed teams use the corresponding ANN10 + DeepONet10. Long stopping requires total, MSE and element L1 plateaus jointly; wall-time stops remain distinct.

Cantera vs NN | Test IC 88 | NRMSE (util.py)

T0 = 1502.1 K P0 = 4.980 atm phi0 = 0.859 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	baseline	0.0536	0.764	5.02	0.609	0.392	0.914	0.0409	1.27	4.48	0	Complete
1 ANN10	soft10k	0.224	1.3	4.59	2.8	0.599	2.9	0.282	449	1.05e+03	0	Complete
1 ANN10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.004 ms**
2 P-only DeepONet10	baseline	0.101	0.92	0.803	0.75	0.101	1.19	0.157	5.76	17.3	0	Complete
2 P-only DeepONet10	soft10k	0.489	1.35	6.09	1	0.775	6.68	0.625	6.29	356	0	Complete
2 P-only DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 8.002 ms**
3 ANN10 + P-only DeepONet10	baseline	0.234	0.143	2.9	0.928	0.832	4.28	0.3	1.4	7.8	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.249	1.3	6.19	3.35	0.612	2.55	0.27	558	5.32e+03	0	Complete
3 ANN10 + P-only DeepONet10	softlong	0.0679	0.808	0.174	3.22	0.368	0.737	0.115	1.72	15.8	0	Complete
4 T/P DeepONet10	baseline	0.0871	0.306	1.15	0.406	0.106	0.998	0.107	2.11	2.76	0	Complete
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	**Stop 0.164 ms**
4 T/P DeepONet10	softlong	0.428	8.19	1.02	1.17	2.59	1	0.99	1.03e+03	3.20e+04	0	Complete
5 ANN10 + T/P DeepONet10	baseline	0.0113	0.0567	0.186	0.0697	0.0111	0.105	0.0136	0.111	0.106	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.0755	1.02	1.11	2.8	0.307	1.05	0.122	3.37	7.39	0	Complete
5 ANN10 + T/P DeepONet10	softlong	0.0654	1.01	0.962	5.33	0.983	1.01	0.162	211	968	0	Complete
6 T/P/ϕ DeepONet10	baseline	0.024	0.107	0.479	0.124	0.0122	0.217	0.0275	0.295	0.232	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.639	8.9	1.05	9.61	2.12	0.0811	1	376	1.67e+04	0	Complete
6 T/P/ϕ DeepONet10	softlong	0.406	4.23	1.17	5.66	0.395	2.85	0.585	0.486	15.9	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	baseline	0.0144	0.076	0.19	0.0787	0.012	0.145	0.0184	0.122	0.129	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.573	6.37	1.08	6.49	1.03	3.68	0.85	2.5	1.25	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	softlong	—	—	—	—	—	—	—	—	—	—	**Stop 0.004 ms**

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
Baseline/soft-short:10000 updates. Soft-long continues the saved10k states at mu10. Each mixed stage uses the corresponding ANN10 and DeepONet10.
All columns compare the NN curve with the same IC Cantera curve. Every graph row includes baseline10k, soft10k and soft-long.